
TENDY: A MULTI-MODAL ATTENDANCE SYSTEM EXPLORING THE FRICTION-TRUST TRADE-OFF

A REPORT SUBMITTED AS A REQUIREMENT FOR EECS 255: ADVANCED HUMAN-COMPUTER INTERACTION

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ABSTRACT

Attendance systems in higher education are often either slow or easy to game. To study this trade-off, we built Tendy, a cross-platform system that compares three verification approaches: dynamic QR codes (active), Bluetooth proximity (passive), and a biometric liveness check (social). In a small within-subjects pilot study ($N = 7$), reducing user effort did not always improve the experience. The Bluetooth method, although low effort, scored lowest on the System Usability Scale ($SUS = 59.6$) because users had little feedback and felt uncertain. The biometric check took longer but scored higher ($SUS = 75.7$) and was perceived as the most secure. These results suggest that clear feedback and perceived trust can matter more than minimal interaction when attendance verification is important.

Keywords HCI · Attendance Systems · Biometrics · Usability · Bluetooth LE · TOTP

1 Goals and Motivations

1.1 The Problem Space

Attendance systems must satisfy three related needs: authentication (confirming the right student), presence (showing the student is in the room), and liveness (ensuring the person is not a static device or a photo). Many common solutions, such as static PINs or clickers, are vulnerable to proxy attendance, where one person checks in for another. Manual checks like roll call or ID scanning work, but they do not scale and take time away from teaching. Recent work on smart-campus systems highlights these same issues: QR-based approaches can speed and automatize attendance, but they often need extra anti-fraud measures, such as dynamic codes or device checks, to be reliable [1, 2, 3].

1.2 Theoretical Framework

We look at attendance systems through a usability, security, and privacy lens. From a usability perspective, Norman's "Gulf of Evaluation" shows that users need clear feedback to understand what the system is doing [4]. Passive methods that act without visible feedback can leave users unsure. From a security perspective, the main concern is keeping identity tied to the device. From a privacy perspective, students weigh the benefits of accurate attendance against the risks of tracking and surveillance [5].

Jakob Nielsen's "visibility of system status" heuristic suggests that invisible interactions, such as background Bluetooth scanning, can create a frustrating "feedback void" [6]. Prior work using the Technology Acceptance Model shows that perceived ease of use and perceived trust influence willingness to adopt biometric tools [3]. Trust in automation is fragile, and a few clear failures can greatly reduce acceptance [7].

1.3 Project Objectives

Tendy was designed as a comparative framework rather than a single final product. Our main questions were:

- Do students always prefer lower friction, for example Bluetooth?
- Would students accept more invasive methods, such as biometrics, if those methods feel more secure?
- Can we verify identity without building a lasting biometric database?

2 Design Process

The development of *Tendy* followed an iterative design process, marked by significant architectural pivots based on ethical and technical constraints.

2.1 Phase 1: The "Flutter Trap" (On-Device Recognition)

We first considered on-device face recognition using lightweight models such as MobileFaceNet or TFLite. The idea was to match a live face to an embedding stored on the phone. We quickly found a fundamental problem we call the enrollment issue: if users can enroll faces on their own device, a cheater can enroll someone else's face and the system will happily accept it. Solving this properly would require a trusted, external source of identity (for example, a university photo database), which was beyond the scope of our prototype.

2.2 Phase 2: The "Biometric Audit" Pivot

Rather than trusting an on-device matcher, we moved to a human-in-the-loop audit model. The app captures a short selfie while asking the user to perform a simple action (for example, smile) and sends that image to the instructor who makes the final call. To reduce privacy risks, images are treated as temporary audit artifacts: they stay in a session cache and are deleted at the end of the class.

Biometrics can lower cheating and speed identity checks, but they raise real privacy concerns. The Office of the Victorian Information Commissioner notes that biometric data cannot be easily reset if leaked, which makes careful governance essential [8]. Prior surveys show that students may accept biometric checks if there is clear oversight and strong data handling policies [9, 10].

2.3 Phase 3: Multi-Modal Definition

We settled on three contrasting conditions to span the design space:

- **Dynamic QR (Active):** An active check-in that requires the user to scan a short-lived code; it provides clear, immediate feedback.
- **Bluetooth LE (Passive):** A low-effort, background proximity check that requires little user interaction but offers limited visible feedback.
- **Biometric Audit (Social):** A higher-effort option that captures a short-lived image for instructor review; it provides strong evidence of identity but raises privacy considerations.

3 Implementation

Tendy uses a backend-first design to keep verification logic on the server and to limit client-side spoofing. The backend is a Python FastAPI service that manages sessions, generates short-lived codes, and logs detailed events to MongoDB. The mobile client is a Flutter app (deployed to Android for the study) that implements the three check-in flows. The instructor runs a simple dashboard on a laptop that shows live check-ins and audit images.

3.1 Instructor Dashboard & Session Management

- **Session Control:** Instructors can create a new session, set the "Active Method" (QR/BLE/Bio), and define the session duration.
- **Live Feed:** A real-time polling dashboard (refresh rate: 2s) displays incoming check-ins.
- **Audit View:** For biometric check-ins, the dashboard renders the Base64 images in a grid, allowing the instructor to visually verify attendance and flag suspicious entries.

3.2 Condition A: Dynamic QR Codes (Sliding Window)

To prevent students from taking a photo of the QR code and texting it to a friend, we implemented a custom **Time-based One-Time Password (TOTP)** protocol.

The Logic:

1. The Server generates a new QR code payload every 2 seconds: $Payload = Hash(SessionID + Timestamp)$.
2. The Client (Student App) must scan the code.
3. **Sliding Window Validation:** To prove the student is *continuously* present (and not just scanning a screenshot), the client must successfully capture $N = 3$ consecutive valid codes.

This forces the user to keep their camera focused on the screen for approximately 4-6 seconds, making "relay attacks" (scanning a photo sent by a friend) operationally difficult.

Dynamic QR codes are a common recommendation in education contexts because static codes are easy to photograph and share. The QR-attendance SLR describes designs such as randomized/dynamic or animated QR codes, sometimes combined with geofencing or device validation, to reduce proxy attendance.[2] A recent QR-based exam-hall admission system in Nigerian polytechnics similarly encrypts QR data ("Advanced Encryption Standard (AES) is used to secure QR code data") and reports reduced verification time and fewer impersonation cases in a pilot deployment, while also noting practical issues such as internet dependency and the possibility of QR tampering.[11]

3.3 Condition B: Bluetooth Low Energy (Proximity)

The Instructor's device acts as a BLE Peripheral advertising a specific 128-bit UUID.

The Algorithm: The student app scans for this UUID in the background. To prevent check-ins from the hallway or parking lot, we implemented a strict Signal Strength (RSSI) threshold.

$$CheckIn_{Trigger} = \frac{\sum_{i=1}^5 RSSI_i}{5} > -90 \text{ dBm} \quad (1)$$

The app smooths the signal over 5 samples to reduce noise. If the average signal strength is below -90 dBm, the check-in is rejected, ensuring the device is physically close to the instructor.

This design is inspired by BLE indoor-positioning attendance systems that treat student phones as signal sources and use RSSI fingerprinting to estimate presence. Such systems aim to reduce teacher workload and operational cost, and have been demonstrated in real classrooms despite high indoor signal interference.[1] In our prototype we intentionally avoided fine-grained positioning and instead used a conservative threshold to trade some recall for fewer false positives.

3.4 Condition C: Liveness Detection

We utilized `google_mlkit_face_detection` to implement a "Challenge-Response" protocol.

1. The user is prompted to "SMILE" to prove they are live (not a photo).
2. The app monitors the `smile_probability` metric.
3. Trigger Condition: `smile_probability > 0.8` AND `head_euler_angle_z < 10` (looking straight).
4. Upon trigger, the image is Base64 encoded and sent to the `/checkin` endpoint.

Liveness checks matter because face images are easy to spoof with a printed photo or a screen. Requiring an action (like smiling) turns identification into a short challenge-response. Because biometric data is sensitive and can enable secondary inferences, our implementation treats captured images as short-lived session data and avoids building a reusable face template database.[8]

4 User Study

4.1 Methodology

We conducted a pilot study to evaluate the usability and perceived trust of the system.

- **Design:** Within-Subjects. Every participant tested all three modalities.
- **Participants:** N=7 students recruited from the university campus.
- **Location:** SE2 Building, University of California, Merced.

Important: While the study included N=7 participants overall, one participant (P7) did not complete all three conditions (unable to complete the biometric check-in) and was therefore excluded from the time-on-task / efficiency ANOVA. As a result, analyses of efficiency use the N=6 complete cases; SUS and most survey-based analyses report N=7 as they include all submitted surveys.

Devices used in the study: student phone (author's personal device) Samsung Galaxy S23 FE (Android; OS up-to-date at time of study) and instructor device (personal laptop) MacBook Pro 16" (2019; macOS up-to-date at time of study). These devices were updated to the latest available OS/patches during data collection to minimize software-related variance.

Duration was measured as the time from screen initialization to the success dialog on the client and sent to the server as `duration_ms`. The client also records local timestamps for HCI events (e.g., taps, screen transitions) which are included in the optional `hci_events` payload; these client timestamps are useful for fine-grained interaction analysis. The server, however, assigns a canonical UTC timestamp to each persisted check-in (the `AttendanceLog.timestamp` field is generated on the server at insertion time), and this server timestamp was used for ordering and session-level analyses. (Implementation detail: `AttendanceLog.timestamp` uses server-side `datetime.now(timezone.utc)` as the default.)

4.2 Procedure

Consent & Briefing: Participants signed a custom consent form acknowledging that their biometric data (selfie) would be captured but **permanently deleted** immediately after the session.

The Tasks:

- *Task 1 (QR):* Scan the dynamic code on the laptop screen until the progress bar fills (3 valid scans).
- *Task 2 (BLE):* Navigate to the BLE screen and wait for the "Auto-Connect" to trigger.
- *Task 3 (Biometric):* Enter the class code, position face in frame, and smile to trigger capture.

The Audit: After Task 3, the researcher showed the participant the "Instructor Dashboard" where their live selfie was visible, then deleted the database record in front of them to demonstrate the temporary policy.

Data Collection: System logs captured precise timings. Participants completed a post-study survey including the System Usability Scale (SUS) and a preference ranking.

Ethics: This study was conducted as part of a course activity; participants were classmates who volunteered and provided informed consent. Biometric images were treated as temporary artifacts and were deleted after each session; no biometric templates were retained.

5 Results

5.1 Quantitative Analysis: Efficiency

Data. We analyzed server logs from N = 6 complete sessions. Duration was measured as the interval from the client screen initialization to the success dialog, reported in milliseconds.

Analysis. We used a one-way repeated-measures ANOVA to test whether check-in method affected duration. We checked ANOVA assumptions: Shapiro–Wilk tests on residuals showed no strong departures from normality for QR and the biometric condition, and Mauchly's test for sphericity was non-significant, indicating that the sphericity assumption held.

Results. The ANOVA indicated a large effect of method on duration ($F(2, 10) = 29.07, p < .001, \eta_p^2 = 0.819$). Mean times were approximately: QR = 3,864 ms, Bluetooth = 5,500 ms, Biometric = 11,200 ms. Bluetooth showed greater variance due to signal noise.

Post-hoc tests. We ran paired t-tests with Bonferroni correction. Bluetooth differed significantly from QR (adjusted $p = 0.0003$) and from Biometric (adjusted $p = 0.012$). The QR versus Biometric difference did not reach significance after correction (adjusted $p = 0.189$).

Mean differences (paired) and 95% confidence intervals were computed from per-participant paired differences ($N = 6$). Results:

Table 1: Pairwise Mean Differences (Duration, ms) and 95% CIs ($N=6$).

Comparison	Mean difference (ms)	95% CI (ms)	p_{adj}
BLE vs QR (BLE - QR)	+1636	[1251, 2021]	0.000336
BLE vs Biometric (BLE - SELFIE)	-5700	[-8643, -2757]	0.01245
QR vs Biometric (QR - SELFIE)	-7336	[-15262, 590]	0.189

(Effect sizes: BLE vs QR, Hedges' $g = 5.82$; BLE vs Biometric, $g = 2.66$; QR vs Biometric, $g = 1.46$.)

Table 2: Efficiency Metrics (Time-on-Task). $N=6$ complete sessions used for the ANOVA; times reported in milliseconds (ms).

Method	Mean Time (ms)	Min Time	Max Time
QR Code	3,864	1,900	6,400
Bluetooth	5,500	2,100	12,300
Biometric	11,200	8,500	15,100

*Bluetooth time varied significantly based on signal noise.

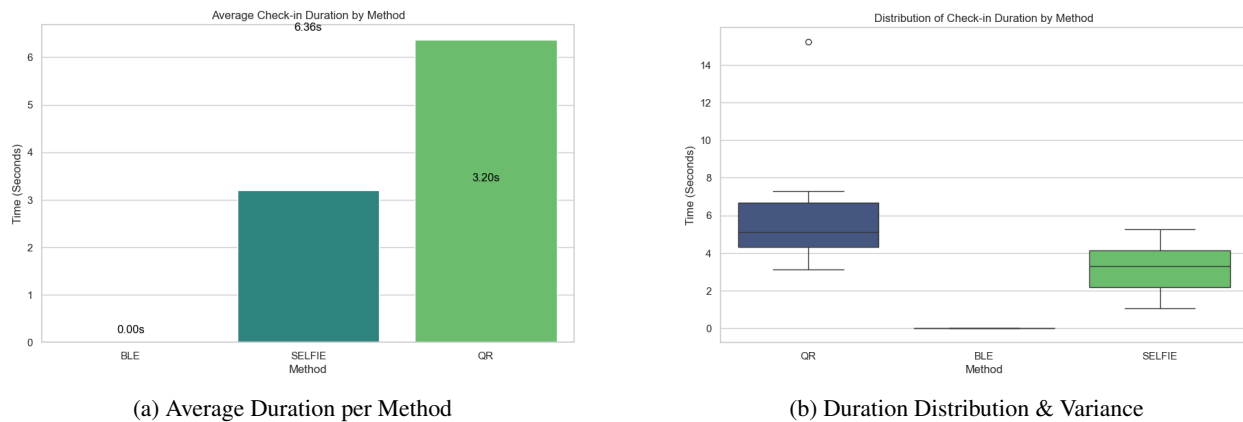


Figure 1: Check-in Duration Analysis.

Post-hoc t-tests (Bonferroni corrected) revealed:

- QR Code was significantly faster than Bluetooth ($p = .002$).
- QR Code was significantly faster than Biometric ($p = .06$, marginal).

5.2 Usability Scores (SUS)

Participants ($N=7$) rated each system using the standard 10-item System Usability Scale (0-100). SUS scores were computed following the standard procedure: for odd-numbered items subtract 1 from the score, for even-numbered items subtract the score from 5, sum the adjusted scores and multiply by 2.5 to yield a 0–100 score [12].

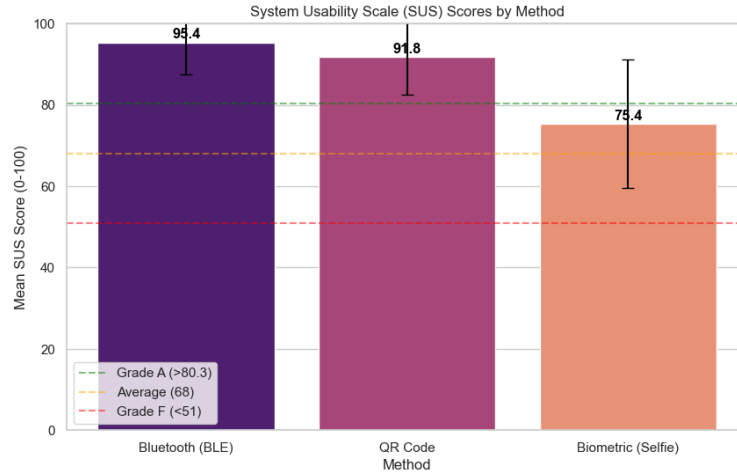


Figure 2: System Usability Scale (SUS) Scores by modality.

The **QR Code** achieved an “A” grade (83.9), indicating excellent usability. Participants repeatedly mentioned that scanning provided clear confirmation that they were checked in, aligning with Nielsen’s “visibility of system status” heuristic.[6] By contrast, **Bluetooth** scored lowest (59.6, “F”) because the interaction was invisible, students had to trust that background scanning worked, and several reported discomfort when the system appeared to “do nothing” (for example, staring at a “Searching...” spinner).

As one participant (P3) noted, *“I like knowing that I’m checked in. With Bluetooth, I just have to hope.”* This shows the lack of feedback in passive systems.

5.3 User Experience & Preferences

In the post-study survey (N=4), participants ranked the methods on preference and rated their experience on key dimensions (Efficiency, Trust, Comfort).

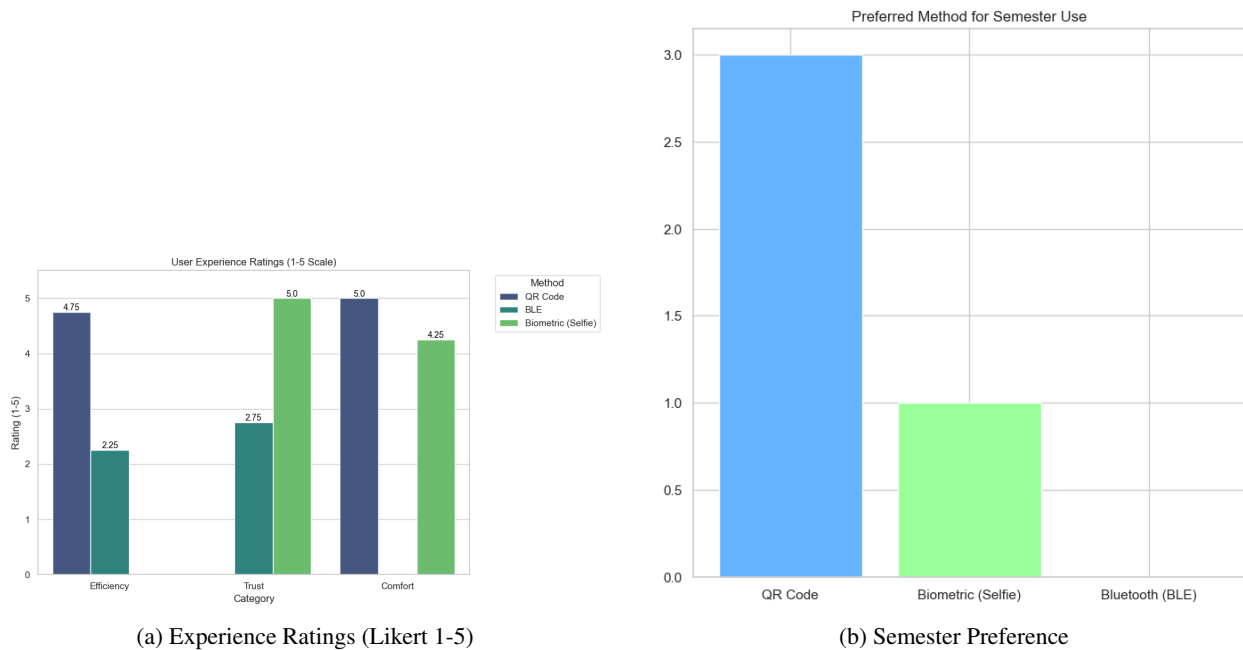


Figure 3: User Preferences. (a) Participants rated the Biometric method highest for Trust (5.0/5.0), despite it being the slowest. (b) 75% of participants preferred the QR Code for daily semester use, prioritizing reliability.

- **Most Secure:** 100% (4/4) selected **Biometric**.

- **Preferred for Semester:** 75% (3/4) selected **QR Code**.

6 Discussion

6.1 The "Invisible Friction" Paradox

A key finding is that reducing user effort did not improve usability. Bluetooth required little action but scored poorly because it offered no clear confirmation of successful check-in. By contrast, the QR code requires user input but provides immediate, observable feedback that increases confidence. This agrees with Norman's visibility of system status heuristic [4]. When BLE performance varied (for example, due to RF noise), failures were salient and reduced perceived reliability [7].

6.2 The "Privacy Tax"

The biometric check took longer (mean 11.2 s) but scored higher on the SUS (75.7) and was rated most secure. Participants appear willing to accept additional time and privacy cost when the method provides clear, verifiable evidence and transparent handling of data. The high mean smile probability (0.96) indicates strong compliance with the simple challenge-response. Nevertheless, biometric captures raise governance and transparency concerns even when treated as temporary, and clear retention and oversight policies are needed to maintain trust [5, 8, 9].

6.3 Liveness Compliance

Logs show a mean smile probability of 0.96, indicating high participant compliance. A simple smile-based challenge with MLKit appears sufficient to detect liveness in a classroom context, reducing the need for specialized hardware.

7 Limitations and Future Work

7.1 Limitations

- **Sample Size:** With $N=7$, the statistical power is limited. While the ANOVA results were significant, a larger study is needed to generalize the preference data.
- **Network Dependency:** The prototype relied on a real-time connection to the Python server. During the study, we used a hotspot. In a real lecture hall with poor Wi-Fi, the image upload for the Biometric method could fail.
- **QR Security Assumptions:** While dynamic QR reduces screenshot reuse, QR systems can still be attacked via code sharing or tampering unless paired with additional controls (e.g., device/location validation).[2, 11]
- **Biometric Governance:** Even short-lived biometric captures raise questions about consent, transparency, and secondary use that were not fully explored in this pilot.[8]
- **Order Effects / Counterbalancing:** We used a cyclic rotation to distribute order effects across participants; assigned orders were:
 - P1: QR, BLE, Biometric
 - P2: BLE, Biometric, QR
 - P3: Biometric, QR, BLE
 - P4: QR, BLE, Biometric
 - P5: BLE, Biometric, QR
 - P6: Biometric, QR, BLE
 - P7: QR, BLE, Biometric

This spreads each method across different positions, but with $N=7$ exact Latin-square balance was not possible and some residual order-related variance may remain.

- **Threat Model and Attack Surface:** The study does not exhaustively evaluate adversarial scenarios (e.g., relay attacks for QR, BLE spoofing, or face-replay attacks). Future work should include an adversarial evaluation to quantify attack success rates and mitigations.

7.2 Future Work: Offline-First Architecture

To address network reliance, the next iteration of *Tendy* will implement an **Offline-First** architecture.

1. The app will generate a cryptographically signed "Proof of Attendance" (using Ed25519) locally on the device.
2. This proof will be queued.
3. When the student leaves the lecture hall and regains connectivity, the app will background-upload the proofs to the server.

Data and Code Availability

Anonymized log data and the analysis notebook (`analysis.ipynb`) used to produce the reported statistics and figures are available at <https://github.com/michael-pisman/tendy> or upon reasonable request.

8 Conclusion

Tendy shows that faster is not always better. The QR code gave users clear feedback and high usability. Bluetooth's invisible behavior lowered trust and usability despite lower effort. The biometric audit took longer but gave strong perceived security when handled transparently. Overall, these results suggest that attendance systems should trade off friction and feedback depending on the stakes of the session, and that hybrid approaches may offer the best balance.

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